IV. Results

(S)ARIMA on Smartwatch Sleep Tracking data

```
sleep <- read_csv("C:/Users/prasa/Desktop/data-science/RStudio/pstat174/smartwatch_sleep_dataset.csv")
```

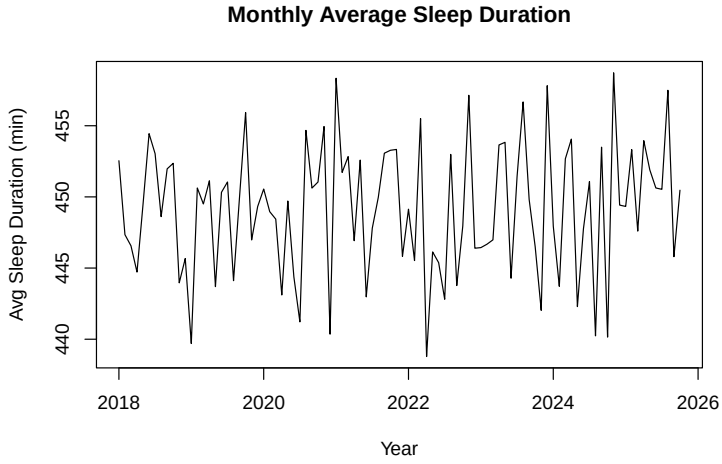
```
## Rows: 20000 Columns: 45
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (5): user_id, daily_label, timezone, gender, device_model
## dbl  (36): duration_minutes, sleep_latency_minutes, wake_after_sleep_onset_m...
## dttm  (3): sleep_start_timestamp, sleep_end_timestamp, created_at
## date  (1): date_recorded
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# avg sleep duration per month
sleep_monthly <- sleep %>%
  mutate(month = floor_date(date_recorded, "month")) %>%
  group_by(month) %>%
  summarise(
    avg_sleep = mean(duration_minutes, na.rm = TRUE)
  ) %>%
  arrange(month)

start_year  <- year(min(sleep_monthly$month))
start_month <- month(min(sleep_monthly$month))

sleep_ts <- ts(
  sleep_monthly$avg_sleep,
  frequency = 12,
  start = c(start_year, start_month)
)
```

```
# plot raw time series
plot(sleep_ts,
     main="Monthly Average Sleep Duration",
     xlab = "Year",
     ylab = "Avg Sleep Duration (min)")
```



This shows the monthly average sleep duration from 2018 to 2025. The series oscillates irregularly around a stable level with no systematic upward or downward trend, and the amplitude of fluctuation appears roughly constant over time, suggesting constant variance.

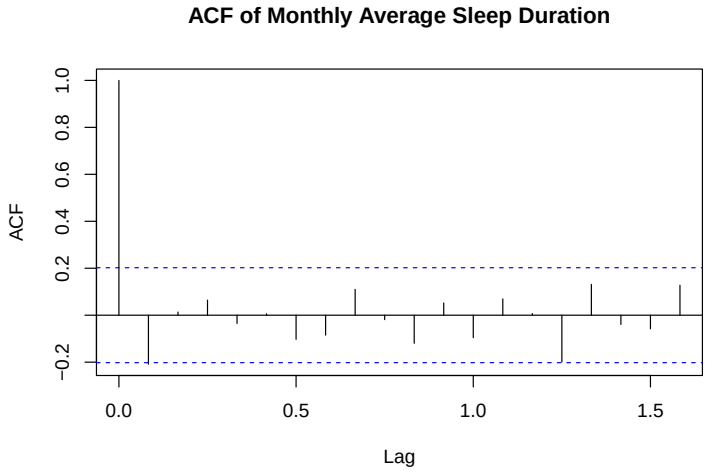
```
adf.test(sleep_ts)
```

```
## Warning in adf.test(sleep_ts): p-value smaller than printed p-value
```

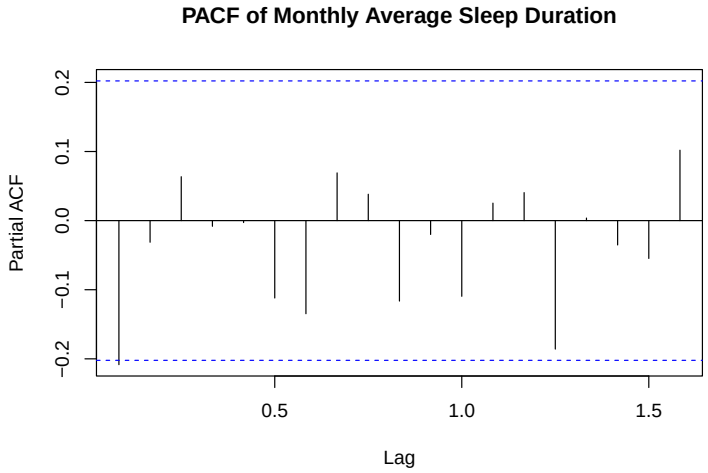
```
##
##  Augmented Dickey-Fuller Test
##
## data:  sleep_ts
## Dickey-Fuller = -4.1722, Lag order = 4, p-value = 0.01
## alternative hypothesis: stationary
```

The plot shows the monthly average sleep duration from 2018 to 2025. The series is stationary (d = 0); no differencing required.

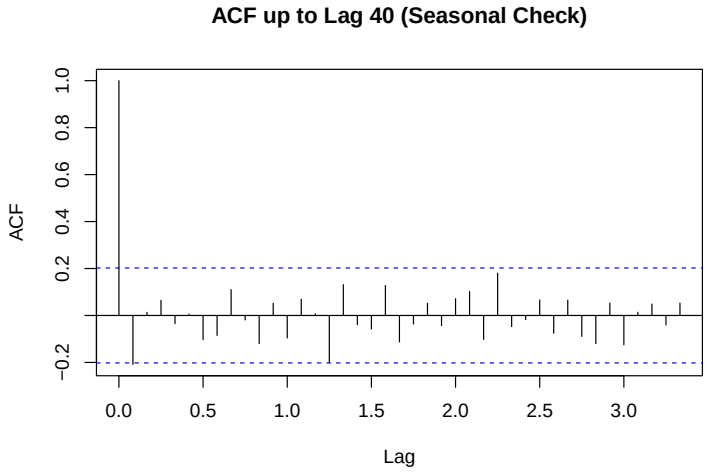
```
acf(sleep_ts, main = "ACF of Monthly Average Sleep Duration")
```



```r
pacf(sleep_ts, main = "PACF of Monthly Average Sleep Duration")
```




```r
acf(sleep_ts, lag.max = 40, main = "ACF up to Lag 40 (Seasonal Check)")
```



ACF plot shows that all lags fall within 95% confidence bands with no significant spikes. PACF shows that all lags fall within 95% confidence bands with no significant spikes. Neither the ACF nor PACF shows any statistically significant autocorrelation beyond lag 0. This means there is no detectable AR or MA structure in the series. No significant spikes visible at lags 12,24,36 which indicates the series does no exhibit monthly seasonality. A SARIMA model was considered but not supported.

Model Selection

```r
# Fit candidate models
fit_000 <- Arima(sleep_ts, order = c(0,0,0))  # white noise baseline
fit_100 <- Arima(sleep_ts, order = c(1,0,0))  # AR(1)
fit_001 <- Arima(sleep_ts, order = c(0,0,1))  # MA(1)
```

```r
# Compare information criteria
AIC(fit_000); AIC(fit_100); AIC(fit_001)
```

```
## [1] 557.8224
```

```
## [1] 555.6628
```

```
## [1] 555.6997
```

```r
BIC(fit_000); BIC(fit_100); BIC(fit_001)
```

```
## [1] 562.909
```

```
## [1] 563.2927
```

```
## [1] 563.3296
```

```r
# Try SARIMA(0,0,0)(1,0,0)[12] - seasonal AR(1)
fit_sarima <- Arima(sleep_ts, order = c(0,0,0), seasonal = list(order = c(1,0,0), period = 12))
summary(fit_sarima)
```

```
## Series: sleep_ts
## ARIMA(0,0,0)(1,0,0)[12] with non-zero mean
##
## Coefficients:
##          sar1      mean
##       -0.1023  449.1180
## s.e.   0.1059    0.4346
##
## sigma^2 = 21.42:  log likelihood = -276.45
## AIC=558.9   AICc=559.16   BIC=566.53
##
## Training set error measures:
##                       ME     RMSE      MAE         MPE      MAPE      MASE
## Training set 0.001326121 4.578291 3.772918 -0.01012209 0.8407021 0.6683618
##                    ACF1
## Training set -0.200929
```

```r
# Also try auto.arima with seasonality allowed
auto.arima(sleep_ts, seasonal = TRUE, stepwise = FALSE, approximation = FALSE)
```

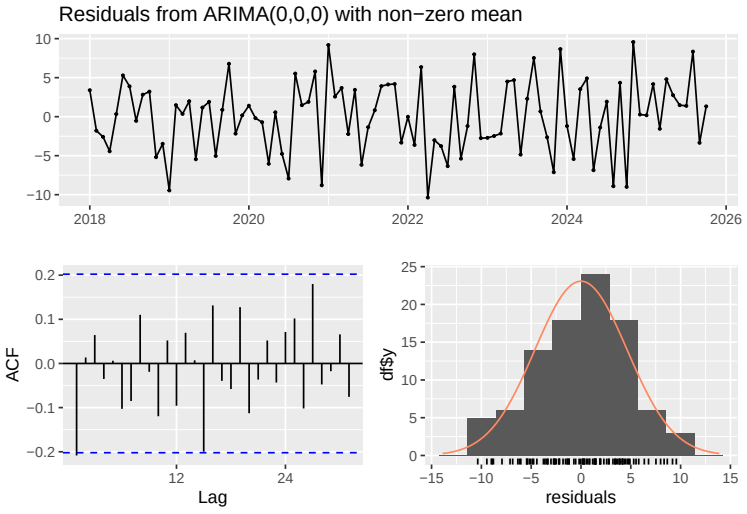
```
## Series: sleep_ts
## ARIMA(1,0,0) with non-zero mean
##
## Coefficients:
##           ar1      mean
##       -0.2076  449.1400
## s.e.   0.1006    0.3852
##
## sigma^2 = 20.71:  log likelihood = -274.83
## AIC=555.66   AICc=555.93   BIC=563.29
```

The results present a nuanced picture. AR(1) achieves the lowest AIC (555.66 vs 557.82 for ARIMA(0,0,0)), suggesting marginally better fit when penalized lightly for the extra parameter. However, ARIMA(0,0,0) achieves the lowest BIC (562.91 vs 563.29 for AR(1)), which applies a stricter penalty of ln(n) approx. 4.56 per parameter versus AIC's penalty of 2. Given n = 96 observations, BIC strongly favors the simpler model. ARIMA(0,0,0) achieves the lowest BIC and is preferred. Alternative models (AR(1), MA(1), SARIMA) provide no meaningful improvement.

```
fit <- Arima(sleep_ts, order = c(0,0,0))
summary(fit)
```

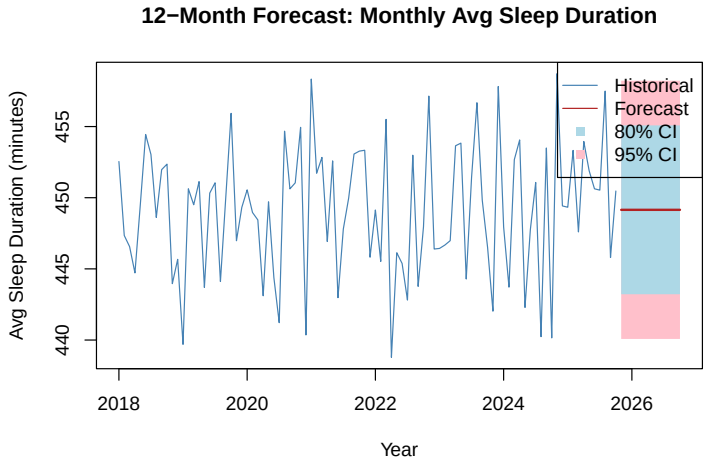
```
## Series: sleep_ts
## ARIMA(0,0,0) with non-zero mean
##
## Coefficients:
##           mean
##       449.1487
## s.e.    0.4749
##
## sigma^2 = 21.42:  log likelihood = -276.91
## AIC=557.82   AICc=557.95   BIC=562.91
##
## Training set error measures:
##                         ME     RMSE      MAE         MPE     MAPE      MASE
## Training set -4.517238e-13 4.603999 3.782252 -0.01052506 0.842813 0.6700153
##                     ACF1
## Training set -0.2084311
```

```
checkresiduals(fit)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,0,0) with non-zero mean
## Q* = 20.397, df = 19, p-value = 0.371
##
## Model df: 0.   Total lags used: 19
```

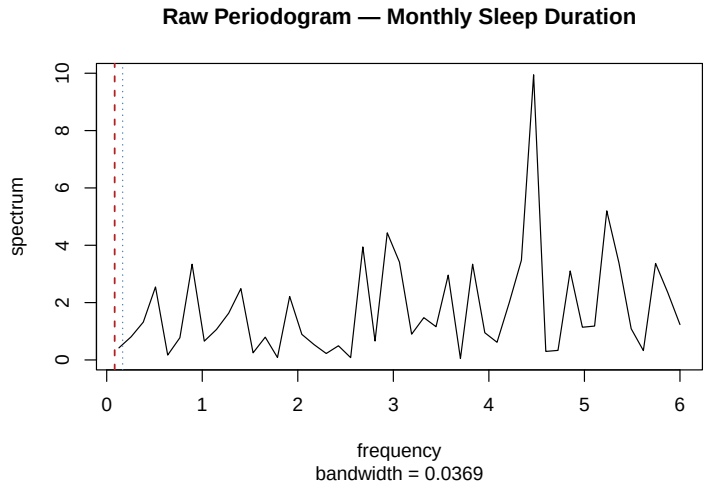
```r
forecast_12 <- forecast(fit, h = 12)
plot(forecast_12,
     main = "12-Month Forecast: Monthly Avg Sleep Duration",
     xlab = "Year",
     ylab = "Avg Sleep Duration (minutes)",
     col  = "steelblue", fcol = "firebrick", shadecols = c("pink", "lightblue"))
legend("topright", legend = c("Historical", "Forecast", "80% CI", "95% CI"),
       col = c("steelblue","firebrick","lightblue","pink"), lty = c(1,1,NA,NA),
       pch = c(NA,NA,15,15))
```



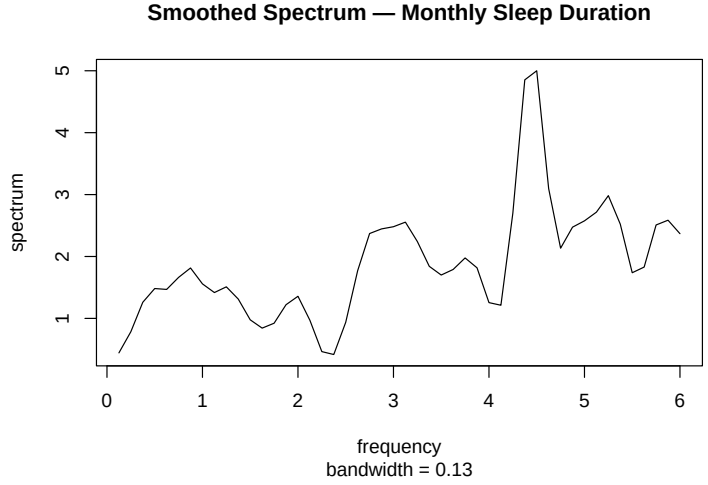
The ARIMA(0,0,0) model estimates mean monthly sleep duration at 449.15 minutes (7 hrs 29 min), with RMSE = 4.60 min and MAE = 3.78 min (MAPE = 0.84%). All three residual diagnostics confirm white noise behavior, and the Ljung–Box test fails to reject the no-autocorrelation null. The 12-month forecast is a flat line at 449.15 min with a constant 95% prediction interval of [440.08, 458.22]; intervals do not widen because the d = 0 white noise structure carries the same variance at every horizon. Overall, monthly sleep duration shows no exploitable temporal pattern; variation around the long-run mean appears to be random noise, likely a product of the dataset's synthetic construction. ARIMA(0,0,0) stands as the most defensible model for this series.

Spectral Analysis on Sleep Data

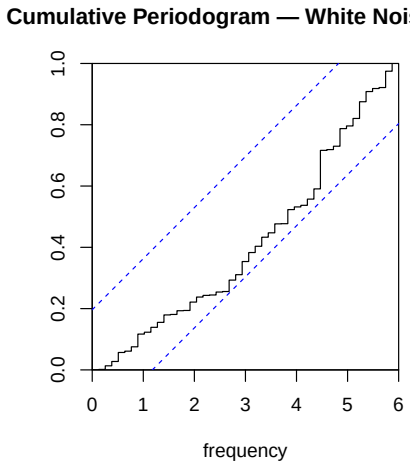
```r
# Raw periodogram diagnostics
raw_spec <- spec.pgram(sleep_ts, log = 'no', taper = 0, fast = FALSE,
                       main = "Raw Periodogram - Monthly Sleep Duration")
abline(v = 1/12, col = 'firebrick', lty = 2, lwd = 1.5)
abline(v = 1/6,  col = 'steelblue', lty = 3, lwd = 1.2)
```




```r
# Smoothed spectrum
smooth_spec <- spec.pgram(sleep_ts, spans = c(3,3), taper = 0.1, log = 'no',
                          main = "Smoothed Spectrum - Monthly Sleep Duration")
```




```r
# Points outside KS bands → reject white noise at that frequency
cpgram(sleep_ts, main = "Cumulative Periodogram - White Noise Test")
```



Overall, no dominant peaks are observed, and there is no evidence of annual or semi-annual cycles. The smoothed spectrum is flat across frequencies. The detected peak occurs at a high-frequency noise region, indicating this is not a meaningful cycle. The cumulative periodogram curve remains within Kolmogorov–Smirnov bounds, thus we fail to reject white noise.

```
# Seasonal power extraction
freqs  <- raw_spec$freq;  powers <- raw_spec$spec
idx_annual     <- which.min(abs(freqs - 1/12))
idx_semiannual <- which.min(abs(freqs - 1/6))
cat('Annual (f=1/12):', round(powers[idx_annual], 2),
    '| Semi-annual:', round(powers[idx_semiannual], 2),
    '| Mean:', round(mean(powers), 2),
    '| Max:', round(max(powers), 2),
    '| Ratio (max/annual):', round(max(powers)/powers[idx_annual], 2))
```

```
## Annual (f=1/12): 0.42 | Semi-annual: 0.42 | Mean: 1.77 | Max: 9.95 | Ratio (max/annual): 23.64
```

Power at annual (1/12) and semi-annual (1/6) frequencies is below the mean spectral level. There is no evidence of seasonal periodicity. Overall spectral diagnostics indicate a flat spectrum with no dominant frequencies, consistent with a white noise process.